

Data Preprocessing Assignment

Assessment weighting: 20% of module marks • Total: 100 marks

Complete four questions (25 marks each) using Python in Jupyter Notebook. The datasets are synthetic teaching resources supplied with this assignment.

Submission and requirements

- Submit one clearly structured .ipynb file in Moodle, with all outputs and plots visible.
- Use methods covered in the four lecture notebooks: pandas, Python string functions, matplotlib, NumPy, scikit-learn scaling, torchvision/PIL, NLTK and VADER.
- Include explanatory markdown and clearly labelled figures. Explain decisions using evidence from your output.
- Do not delete rows merely because they contain missing values without explaining the effect.
- Academic integrity: the assessment must be completed without AI assistance; sign and submit the student declaration of Academic Integrity document in Moodle.

Question 1 — Food inspection data cleaning (25 marks)

1a (5 marks) Load `food_inspections_practice.csv`. Print its dimensions, data types, number of exact duplicate rows and missing/blank values per column. Treat empty strings and whitespace-only values appropriately.

1b (7 marks) Standardise text case and surrounding whitespace; identify and correct inconsistent city spellings; fill missing *AKA Name* values from *DBA Name*. Show before/after frequency tables for City and missing *AKA Name* counts.

1c (5 marks) Remove exact duplicate records using an appropriate method. Investigate whether repeated Site ID records are necessarily duplicates and explain your decision.

1d (5 marks) Visualise the counts of inspection results by Risk category and the distribution of numerical Inspection Score, with informative titles and axis labels.

1e (3 marks) Explain two potential risks associated with (i) deleting inspection records containing missing values and (ii) filling missing *AKA Name* values using *DBA Name*. How might these operations affect the accuracy or interpretation of the cleaned dataset?

Question 2 — Numerical preprocessing and visualisation (25 marks)

2a (5 marks) Load `product_measurements.csv`; inspect missing values and describe the original distributions of Droplet Area and Droplet Count using descriptive statistics and plots.

2b (7 marks) Apply Z-score standardisation and Min-Max normalisation to Droplet Area and Droplet Count. Handle missing values using a method taught in lectures and state what you did.

2c (6 marks) Generate side-by-side distribution plots (histograms or KDE plots) for original, Z-score and Min-Max values for each of the two features. Use clear axes and legends.

2d (4 marks) Calculate or report the minimum, maximum, mean and standard deviation of the two scaled features and interpret the results.

2e (3 marks) Identify extreme observations from the original plots and explain how they affect Min-Max scaled values. Do not remove outliers automatically.

Question 3 — Image transformations and augmentation (25 marks)

3a (5 marks) Load `synthetic_droplets_1.png` and display its original dimensions. Apply `Resize`, `CenterCrop` and `Grayscale`, showing each output and its resulting dimensions.

3b (8 marks) Apply `RandomRotation`, `RandomHorizontalFlip`, `GaussianBlur` and `ColorJitter` using torchvision transforms. Display the original and transformed images in a labelled grid.

3c (5 marks) Apply one random transform at least four times. Explain why repeated calls may give different results and show the outputs.

3d (4 marks) Compare the visual impact of `GaussianBlur` and cropping on apparent droplet boundaries or the visible number of droplets.

3e (3 marks) Explain two reasons why a transformation suitable for general image augmentation may be unsuitable when measuring droplet characteristics.

Question 4 — Text processing and sentiment analysis (25 marks)

4a (5 marks) Load `skincare_reviews.csv`. Define a Python text-cleaning function using regular expressions and lowercasing, and show at least six before/after examples.

4b (7 marks) Tokenise reviews and apply Porter stemming and WordNet lemmatisation separately. Compare outputs for at least five selected reviews, highlighting differences.

4c (7 marks) Use NLTK VADER to calculate compound sentiment scores on original reviews, cleaned reviews and lemmatised reviews. Present the results in a DataFrame and compare their distributions using a suitable plot.

4d (4 marks) Select three reviews where preprocessing changes the score (or, if no changes occur, three with informative unchanged scores). Explain the observed behaviour, including the role of punctuation or negation where relevant.

4e (2 marks) State whether the preprocessing method producing the shortest text is necessarily best for sentiment analysis, with one evidence-based example.